An Unbounded Family of Rational S-Unit Solutions Conjecture 00000000023

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AI-assisted proof; no affiliation claimed

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Abstract

The asserted absolute bound $C|S|^2$ for the number of solutions of $x+y=1$ in rational S-units is false. For every N we use coprime pairs in the square $[1,N]^2$ to obtain distinct solutions $(a/(a+b), b/(a+b))$. At least $N^2/4$ of these pairs are coprime. Taking $N = 2^{2m}$ and using only an elementary primorial bound gives $m|S_N| \le 6N$, so the solution-to-support-square ratio is at least $m^2/144$. This tends to infinity, contradicting every absolute constant. The Lean project checks the actual rational support bridge, not an opaque surrogate predicate.

1 The source assertion

The source names the S-unit equation $x + y = 1$ and asserts the existence of an absolute constant C such that the number of solutions is at most $C|S|^2$. We use the standard rational interpretation: a nonzero reduced rational number is an S-unit when every prime dividing its numerator or denominator lies in the finite prime set S. Supports may be proper subsets of S.

2 The explicit family

For $N \ge 1$ define

$$A_N = \{(a,b) : 1 \le a,b \le N,\ \gcd(a,b) = 1\}, \qquad S_N = \{p \text{ prime} : p \le 2N\}.$$

Map (a,b) to

$$F_N(a,b) = \left(\frac{a}{a+b}, \frac{b}{a+b}\right).$$

Both coordinates are nonzero, add to one, and are reduced because $\gcd(a, a+b) = \gcd(b, a+b) = 1$. All prime divisors of the numerator and denominator are at most $2N$, so the coordinates are genuine rational S_N-units. Uniqueness of reduced positive numerator/denominator representations proves that F_N is injective.

Lemma 1. *For every N, $|A_N| \ge N^2/4$.*

Proof. Every non-coprime pair has a common divisor d with $2 \le d \le N$. The pairs divisible by a fixed d are at most $\lfloor N/d \rfloor^2$. Therefore the bad pairs are bounded by

$$N^2 \sum_{d=2}^{N} \frac{1}{d^2} \le N^2 \left(\frac{1}{4} + \sum_{d=3}^{N} \left(\frac{1}{d-1} - \frac{1}{d}\right)\right) \le \frac{3}{4} N^2.$$

The finite telescoping inequality gives the claim. □

3 Support size and contradiction

For $m \ge 1$, set $N_m = 2^{2m}$, $X_m = 2N_m$, and $S_m = S_{N_m}$. Partition primes up to X_m into those at most 2^m and the rest. There are at most $2^m + 1$ small primes. For the large primes, their product is at least $(2^m)^r$ if there are r of them. The elementary primorial estimate $\prod_{p \le X_m} p \le 4^{X_m}$ follows by partitioning

primes into dyadic intervals and bounding each interval product by a central binomial coefficient. Thus $mr \le 2X_m$. Since $m \le 2^m$,

$$m|S_m| \le m(2^m + 1) + 2X_m \le 2N_m + 4N_m = 6N_m.$$

Here $m(2^m + 1) \le 2(2^m)^2 = 2N_m$, since $m \le 2^m$ and $1 \le 2^m$. These are precisely the definitions and bounds in the Lean development.

Combining this with the injected family gives

$$\frac{|F_{N_m}(A_{N_m})|}{|S_m|^2} \ge \frac{N_m^2/4}{(6N_m/m)^2} = \frac{m^2}{144}.$$

For any proposed absolute C, choose m with $m^2 > 144 \max(C, 0)$. The finite set $F_{N_m}(A_{N_m})$ then contains more than $C|S_m|^2$ actual solutions, so the universal quadratic bound is impossible.

4 Lean correspondence and limits

`RatBridge.lean` defines the actual rational S-unit predicate and proves reduced numerator/denominator and prime-support bridges. `CoprimePairs.lean` proves the all-N coprime-pair lower bound. `PrimeBound.lean` proves the finite prime-support estimate by `primorial_le_four_pow`. `Final.lean` proves injection, actual solution membership, unbounded ratio and the closed negation of a uniform bound. `GeneratedUnits.lean` additionally proves that the constructed witnesses belong to the subgroup generated by -1 and supported primes.

All advertised theorem types and transitive axioms are printed in `Check.lean`. Only `propext`, `Classical.choice`, and `Quot.sound` may occur. No PNT, new axiom, `sorry`, `admit`, `native_decide`, `unsafe` evaluation or hard-coded conclusion is used. The compiler and fixed Mathlib/toolchain remain disclosed trusted infrastructure.

5 Status

This is a complete mathematical disproof under the standard rational S-unit interpretation. It does not claim results for a silently changed notion of S-unit, does not claim human endorsement or upstream acceptance, and does not modify the original conjecture or organizer metadata.